Lecture 24

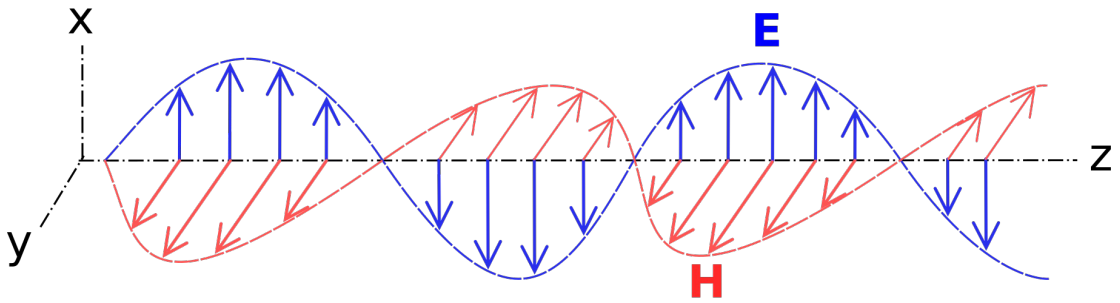
Polarization

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Definition of Polarization

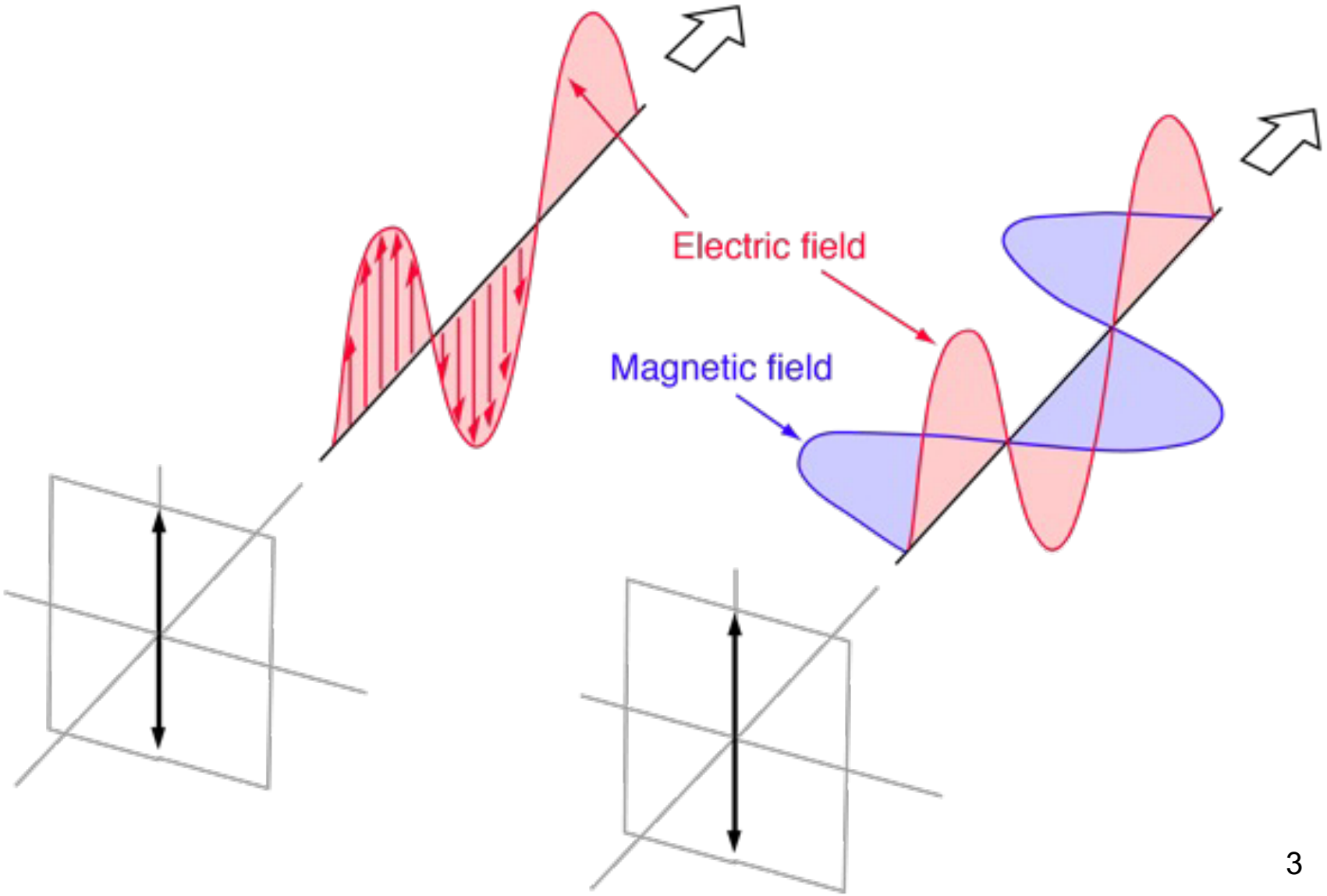
- Describes the "tip" of the time-varying **E** field vector for a particular point in space as it varies in time



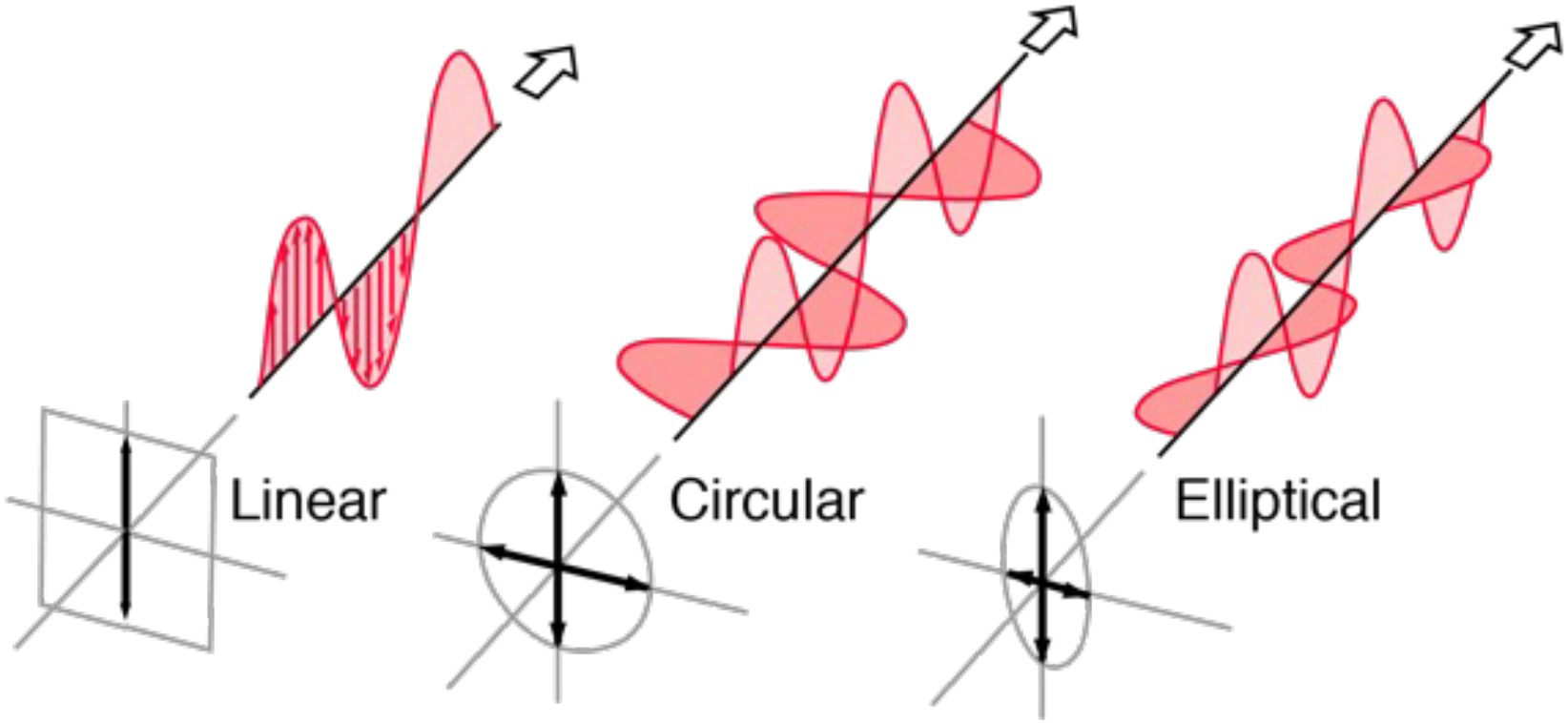
- $\vec{J} = -JS_0 cos(\omega t)\hat{a}_x$ example
  - Field with only one component is always linearly polarized

$$E_x(z,t) = \frac{\eta_0 J_{S0}}{2}\cos(\omega t \mp \beta z)$$

Linear Polarization



Classification of Polarization



How do we treat this mathematically?

Say we have two superimposed fields

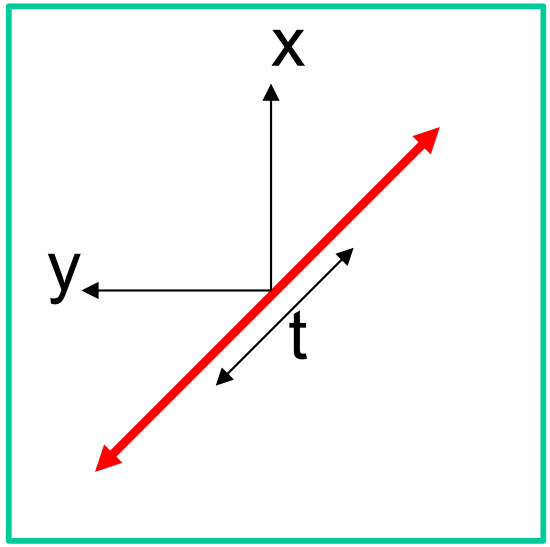
- Propagating in same direction
- Possibly out of phase
- Possibly oriented out of plane with each other

How can we tell if the combined wave will be:

- Linear
- Circular
- Elliptical

Linear Polarization

Plane in space perpendicular to propagation direction



The tip of the E-field vector traces out a straight line

DIRECTION: const.

MAGNITUDE: change w. time.

Linear Polarization

Given two linearly polarized vector fields

$$\vec{E}_1 = E_1 \cos(\omega t + \phi)\hat{a}_x$$

$\mathbf{E}_1$ is horizontal linear polarization (red)

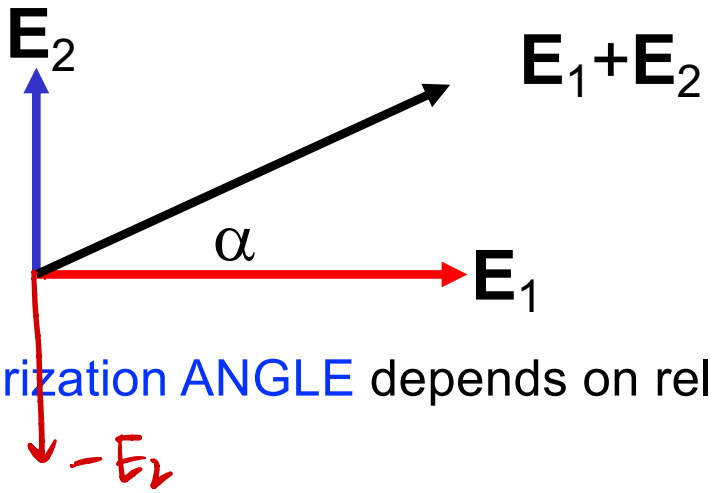
$$\vec{E}_2 = \pm E_2 \cos(\omega t + \phi)\hat{a}_y$$

$\mathbf{E}_2$ is vertical linear polarization (blue)

If the vectors are IN PHASE or 180° apart, $\mathbf{E}_1$+$\mathbf{E}_2$ will also be linear polarization

$$\vec{E}_1 + \vec{E}_2 = \cos(\omega t + \phi)(E_1\hat{a}_x \pm E_2\hat{a}_y)$$

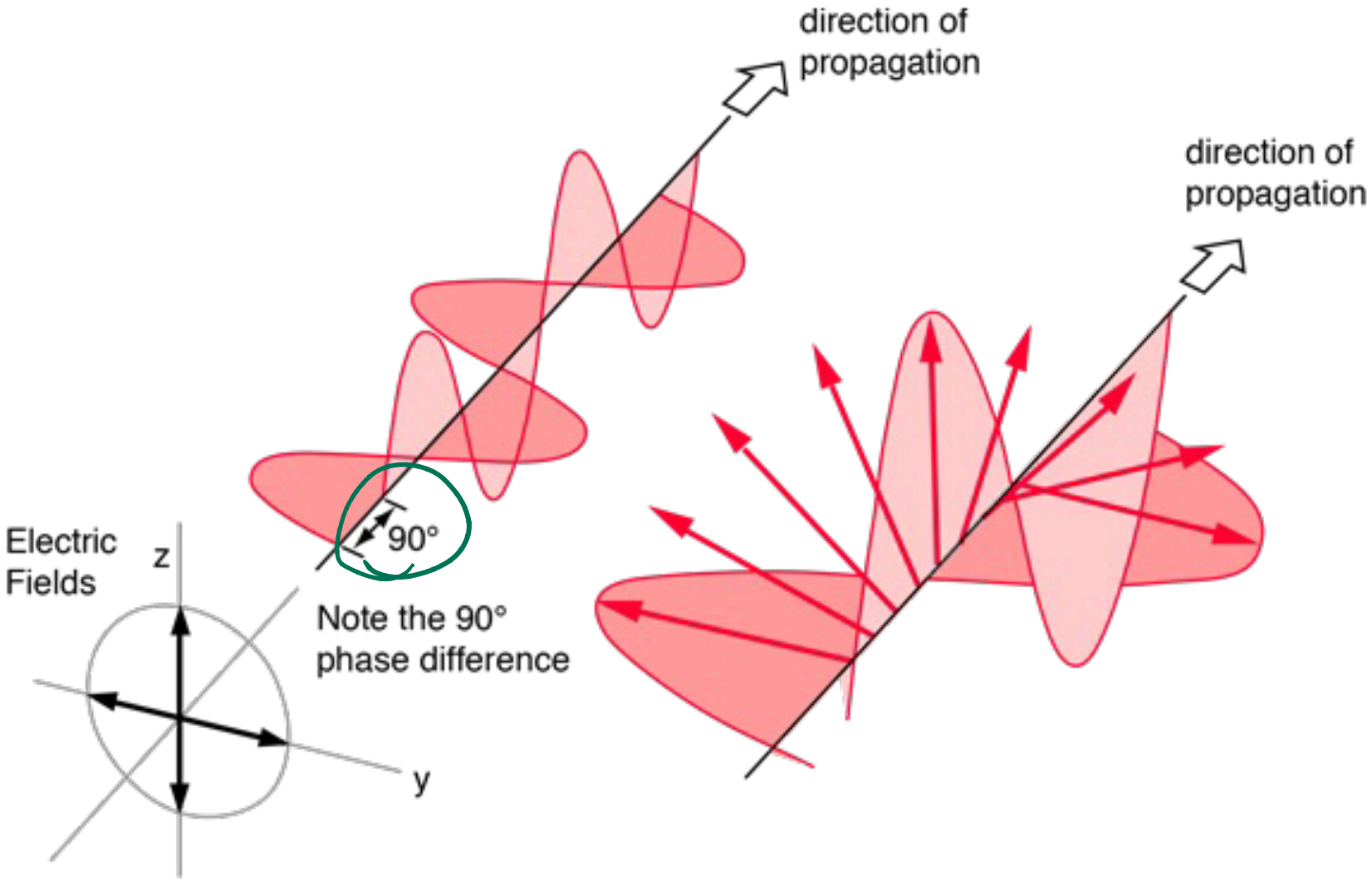
Magnitude changes Direction is constant



$$\alpha = \pm \tan^{-1}\left(\frac{E_2}{E_1}\right)$$

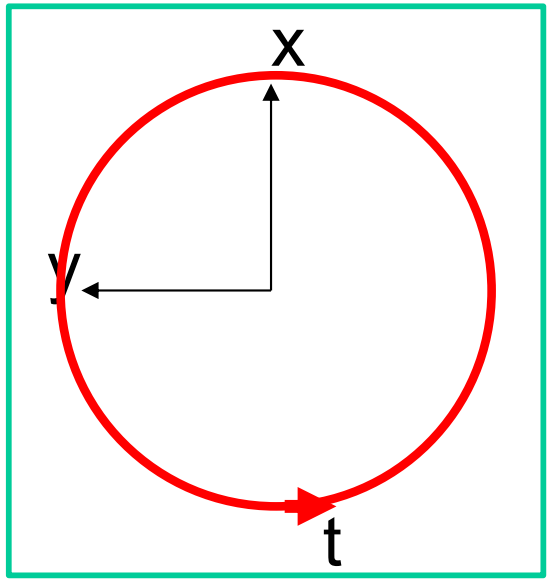
Polarization ANGLE depends on relative magnitudes of E_1 and E_2

Circular Polarization



Circular Polarization

Plane in space perpendicular to propagation direction



The tip of the E-field vector traces out a circle

DIRECTION: change w. time

MAGNITUDE: const.

Circular Polarization

Given two linearly polarized vector fields

$$\vec{E}_1 = E_0 \cos(\omega t + \phi)\hat{a}_x$$ **E**$_1$ is horizontal linear polarization

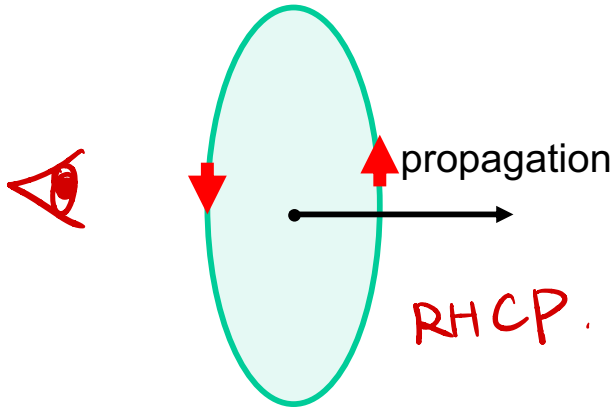
$$\vec{E}_2 = E_0 \sin(\omega t + \phi)\hat{a}_y$$ **E**$_2$ is vertical linear polarization

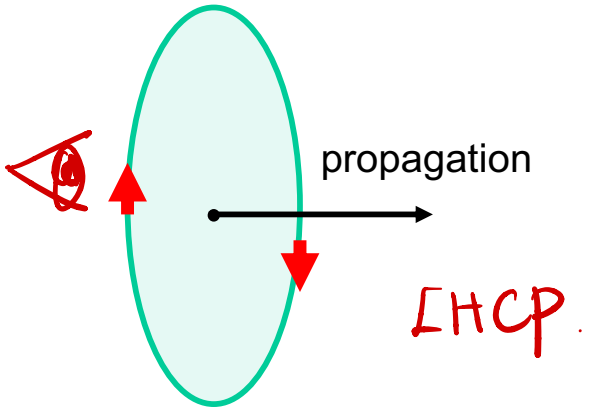
1. ✓ The vectors have EQUAL MAGNITUDES
2. ✓ The vectors are OUT OF PHASE by $\frac{\pi}{2}$
3. ✓ The vectors are PERPENDICULAR $\vec{E}_1 \perp \vec{E}_2$

ALL THREE CONDITIONS MUST BE MET
for **E**$_1$+**E**$_2$ to be circular polarization

How to tell: Left or Right Handed Circular Polarization

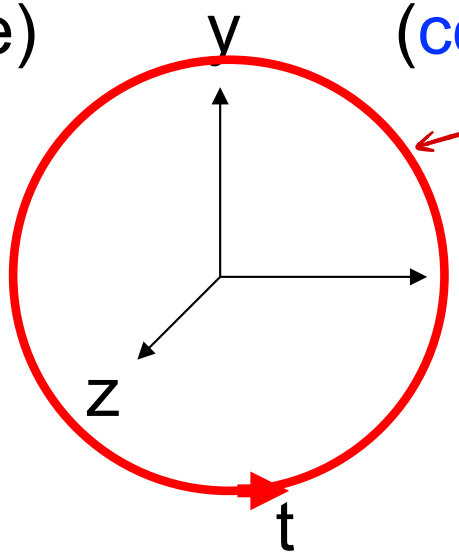
Method 1: Left/right thumb points in **propagation** direction





"Right-Hand" Polarized
(cw as seen by source)

"Left-Hand" Polarized
(ccw as seen by source)



LHCP if moving in $-a_z$

RHCP if moving in $+a_z$

How to tell: Left or Right Handed Circular Polarization

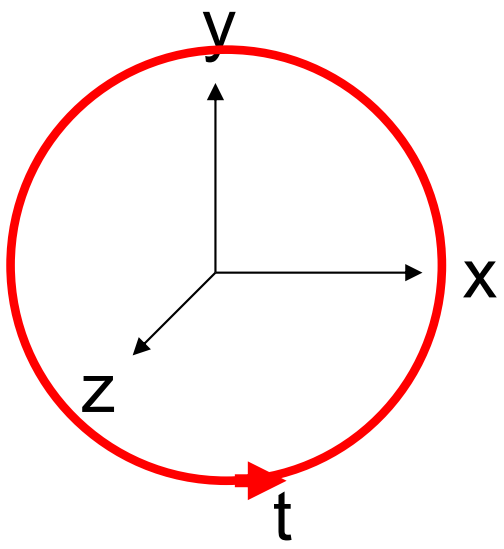
Method 2: It's right handed if
Ahead $\hat{E}$ x Behind $\hat{E}$ is in propagation direction

$$\vec{E}_1 = E_0 \cos(\omega t \pm \beta z)\hat{x} \leftarrow \textit{Ahead}$$

$$\vec{E}_2 = E_0 \sin(\omega t \pm \beta z)\hat{y}$$

$$= E_0 \cos(\omega t \pm \beta z - \frac{\pi}{2})\hat{y} \leftarrow \textit{Behind}$$

It is behind by $\pi/2$ because ωt needs to be $\pi/2$ larger to be at the same part of the wave



$\hat{x}$ $\hat{y}$

$$\hat{E}_1 \times \hat{E}_2 = \hat{z}$$

so RHCP
if moving in $+a_z$

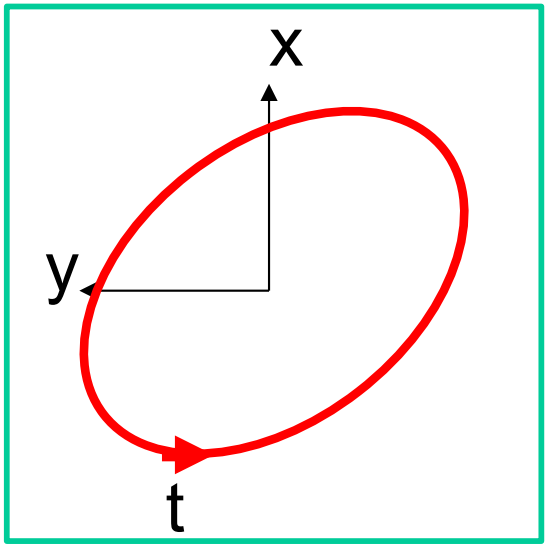
else LHCP
if moving in $-a_z$

$E_1 \neq E_2$.
$\phi \neq 0^\circ, \pm 90^\circ$

Elliptical Polarization

Most general: two combined linearly polarized waves will result in an elliptically polarized wave - if the conditions for linear or circular are not satisfied.

Plane in space perpendicular to propagation direction



The tip of the E-field vector traces out an ellipse

DIRECTION: change. w. time

MAGNITUDE change. w. time

Challenge Question: Forming Circular Polarization

For what value of ϕ, will the following fields add to produce **right**-handed circular polarization:

$$\vec{\boldsymbol{E}}_1 = E_0 \cos(2\pi\times10^8 t + 2\pi z + \frac{\pi}{3})\hat{\boldsymbol{a}}_x$$ — behind.

$$\vec{\boldsymbol{E}}_2 = E_0 \cos(2\pi\times10^8 t + 2\pi z + \phi)\hat{\boldsymbol{a}}_y$$ — ahead

(a) π/3
(b) -π/3
(c) π/6
(d) -π/6
(e) 5π/6

$\vec{S} = -\hat{z}$

$\hat{a}_y \times \hat{a}_x = -\hat{z}$

$\phi - \frac{\pi}{2} = \frac{\pi}{3}$.

$\phi = \frac{5\pi}{6}$.

Two linear P wave → combine → { linear, circular, elliptical

Writing linearly polarized as a sum of circular polarization

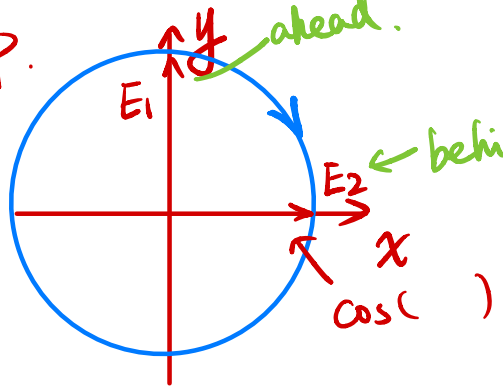
Rewrite linear polarized waves $\mathbf{E} = \mathrm{E}_0 \cos(\omega t + \beta z)\, \hat{a}_x$ as a sum of circular polarized fields.

$-\hat{z}$ wave

$\tilde{E} = E_0 e^{j\beta z}\, \hat{x}$

RHCP LHCP

① RHCP.



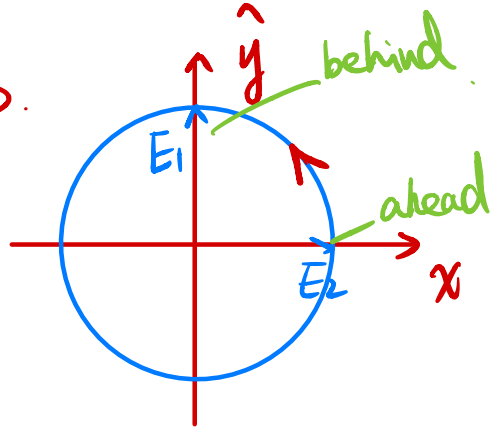
$$\vec{E}_{RHCP} = E_0 \cos(\omega t+\beta z)\,\hat{x} + E_0 \cos(\omega t+\beta z+\tfrac{\pi}{2})\,\hat{y}$$

$$= E_0 \cos(\omega t+\beta z)\,\hat{x} - E_0 \sin(\omega t+\beta z)\,\hat{y}$$

$$\tilde{E}_{RHCP} = E_0 e^{+j\beta z}\,\hat{x} + E_0 e^{j\beta z}\,(j\hat{y})$$

$$= E_0 e^{j\beta z}(\hat{x}+j\hat{y})$$

② LHCP.



$$\tilde{E}_{LHCP} = E_0 \cos(\omega t+\beta z)\,\hat{x} + E_0 \cos(\omega t+\beta z-\tfrac{\pi}{2})\,\hat{y}$$

$$= E_0 \cos(\omega t+\beta z)\,\hat{x} + E_0 \sin(\omega t+\beta z)\,\hat{y}$$

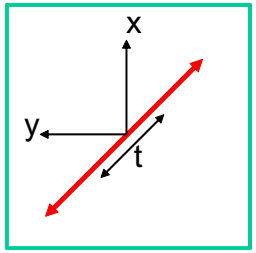
$$\tilde{E}_{LHCP} = E_0 e^{j\beta z}\,\hat{x} + E_0 e^{j\beta z}\,(-j\hat{y})$$

$$= E_0 e^{j\beta z}(\hat{x}-j\hat{y})$$

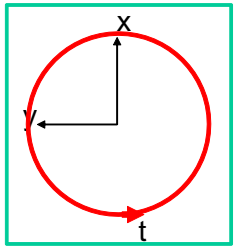
$$E_{linear} = (\tilde{E}_{RHCP} + \tilde{E}_{LHCP})/2$$

Lecture 24 Summary

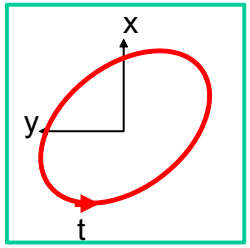
- Polarization describes the "tip" of **E**(t)
- Linear
  - Direction is _____
  - Magnitude _____



- Circular
  - Direction _____
  - Magnitude is _____



- Elliptical
  - Direction _____
  - Magnitude _____



- Next up: Wave reflection